

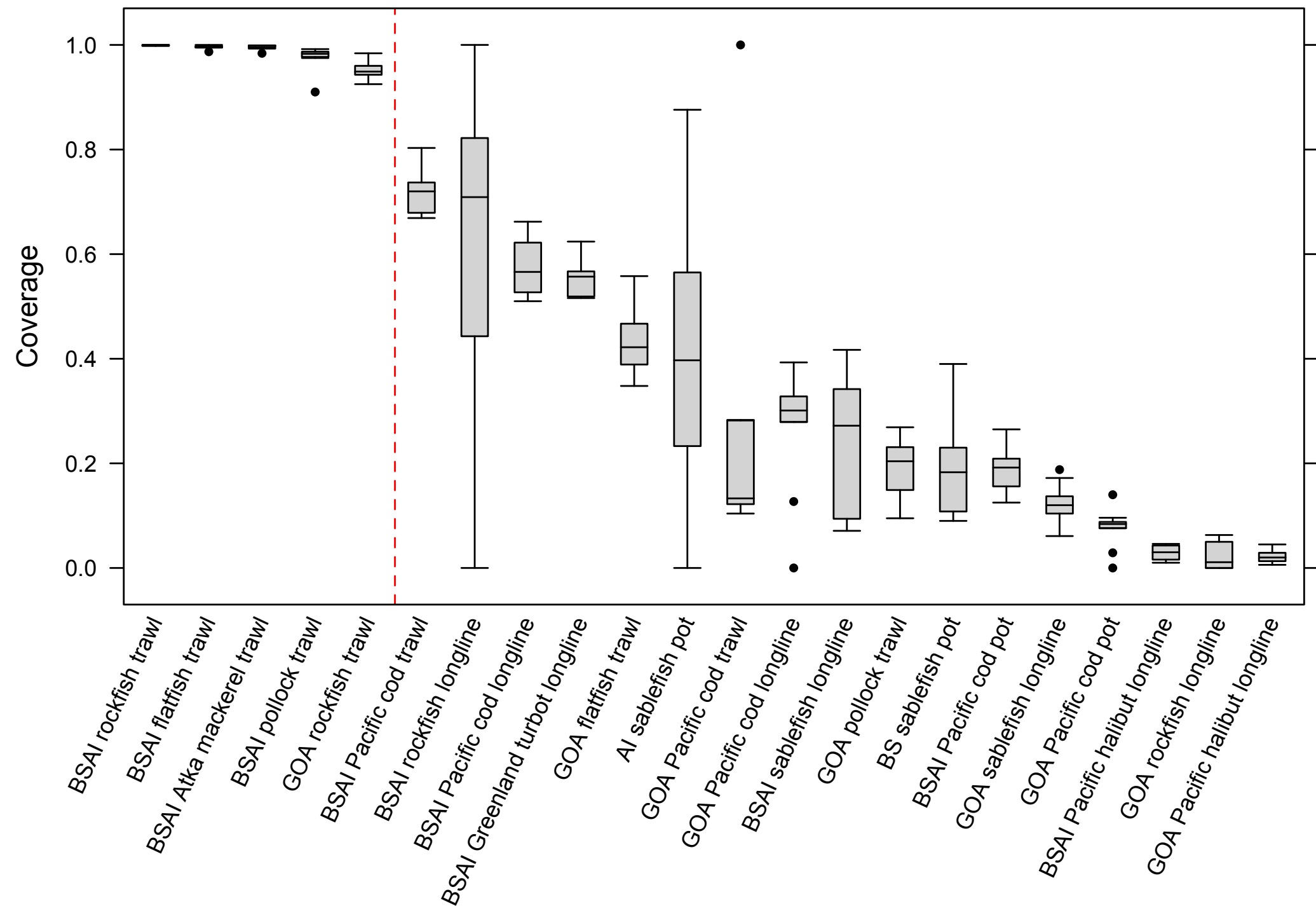
Marine mammal bycatch estimation for Alaska fisheries

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Some basics...

- Bycatch: marine mammals incidentally killed or injured during fishing activities
- Quantifying bycatch required under MMPA
- Rely on data from the North Pacific Observer Program

Observer coverage



Estimating bycatch

- Current method is a ratio estimator (Breiwick 2013)

$$N = rX$$

- Estimates stratified by vessel length, region, and time period
- Combined to obtain annual estimates for each marine mammal species and fishery

Shortcomings

Low coverage and rare events combine to yield:

- Volatile annual estimates (e.g., harbor seals in 2020)
- False zeroes when no bycatch is observed

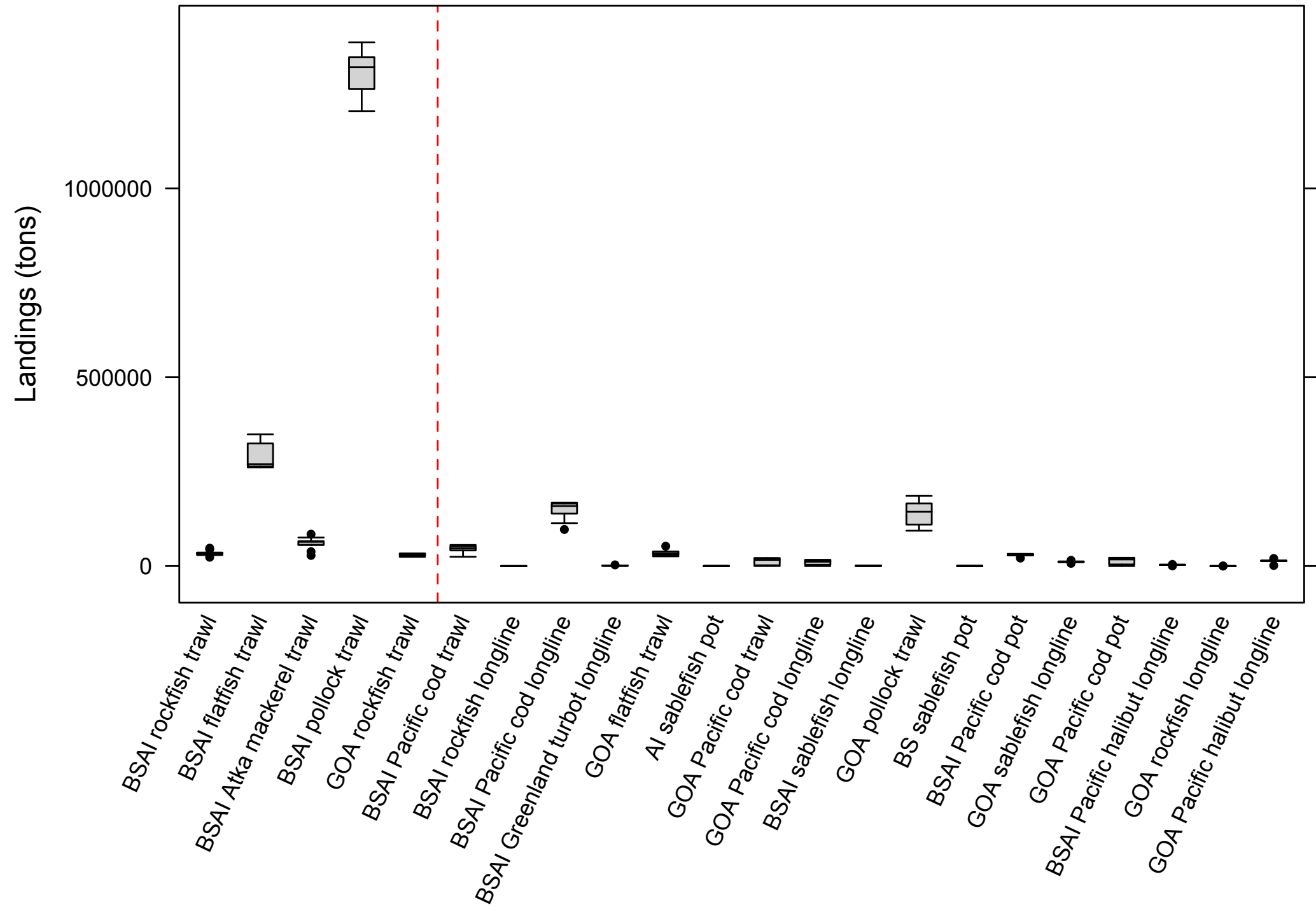
Steller sea lion - BSAI Pacific cod longline

Year	Observer coverage	Observed bycatch	Estimated total bycatch	Variance
2016	0.57	0	0	0
2017	0.58	1	1.58	0.82
2018	0.55	0	0	0
2019	0.52	0	0	0
2020	0.53	0	0	0

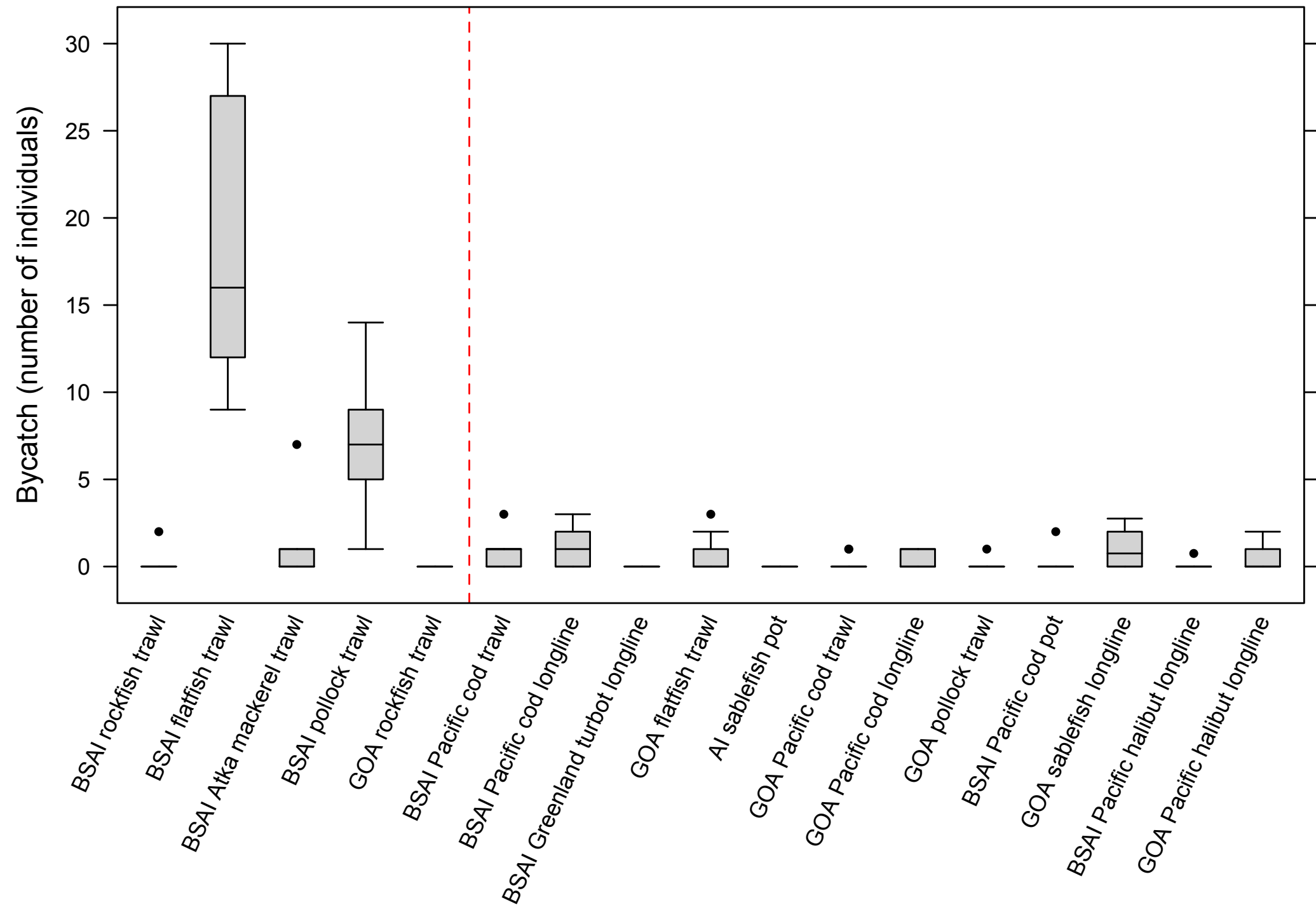
Looking ahead

- Other science centers use ratio estimators, model-based approaches, and machine learning
- Inquire with NWFSC (GLM) and SWFSC (random forest)
- Implement and evaluate alternative method
- Report methods and results in citable format

Landings



Bycatch



$$N = rX, \quad r = \sum_{i=1}^n \left(\frac{y_i}{x_i} \right)$$

$$\text{Var} (N) \propto \sum_{i=1}^n (y_i - rx_i)^2$$